

Build Beautiful Cross-Platform Apps with Uno Platform

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About Me

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Agenda

- What is Uno Platform?
- Develop Your Way – IDE Flexibility
- Accelerate Creation – Live Wizard
- Bridge Design & Code – Figma to XAML / C# Markup
- Visualize Your UI – Hot Design
- Go XAML-Free – C# Markup
- Modernize Legacy Apps – WinUI Migration

What is Uno Platform?

- Open-source framework for building cross-platform apps with .NET
- Single codebase targeting **iOS, Android, macOS, Linux, Windows, and the Web**
- Pixel-perfect UI across all platforms
- 100% native platform capabilities
- Visual design with Uno Platform Studio and Hot Design

Why Uno Platform?

- **Single codebase** — write once, run everywhere
- **Full .NET ecosystem** — NuGet packages, C#, LINQ, async/await
- **Familiar APIs** — WinUI / XAML for UI developers
- **Flexible UI options** — XAML or C# Markup
- **Active community** — open source with strong commercial support

Supported Platforms

Platform	Technology
Windows	WinUI 3 / Windows App SDK (native)
iOS	.NET for iOS (Skia)
Android	.NET for Android (Skia)
macOS	Skia Desktop
Linux	Skia Desktop (X11, Framebuffer)
Web	WebAssembly via Skia

Develop Your Way

Use your favorite IDE on any OS

IDE Choices

- **Visual Studio** (Windows)
 - Full XAML designer & IntelliSense
 - Integrated debugging for all targets
- **VS Code** (Windows / Mac / Linux)
 - Uno Platform extension
 - Lightweight and fast
- **JetBrains Rider** (Windows / Mac / Linux)
 - Rich .NET support
 - Cross-platform debugging

Visual Studio

- XAML Hot Reload for rapid iteration
- Solution Explorer with multi-target awareness
- Integrated emulators and simulators
- NuGet package management

VS Code

- Uno Platform Extension for VS Code
- C# Dev Kit integration
- Terminal-based workflows
- Great for Linux developers

JetBrains Rider

- First-class .NET IDE on all platforms
- XAML preview and editing support
- Powerful refactoring tools
- Integrated unit testing

Accelerate Creation

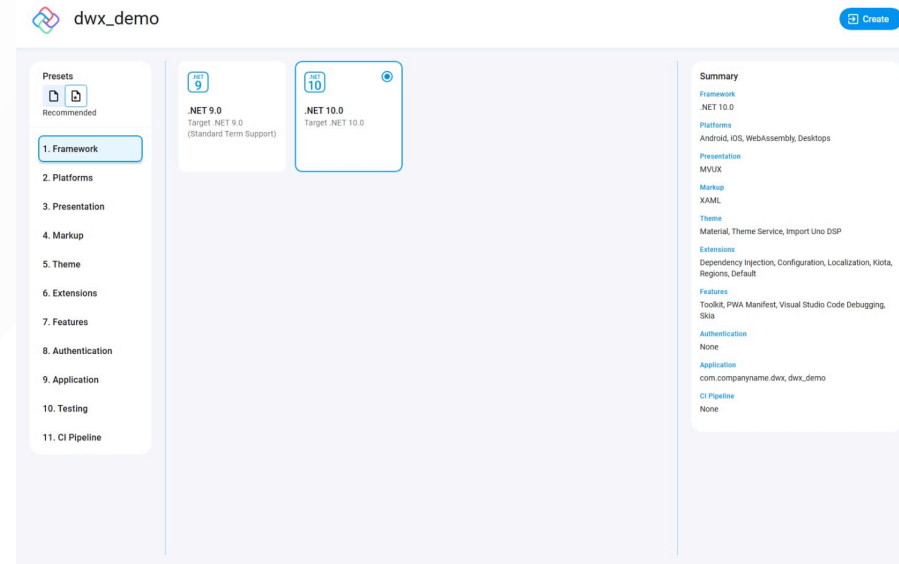
Launch projects instantly with the Template Wizard

Uno Platform Template Wizard

- Web-based project template generator
- Available at **new.platform.uno**
- Choose your target platforms
- Select authentication, navigation, and theming options
- Generates a fully configured solution in seconds

Template Wizard Walkthrough

1. Visit the Uno Template Wizard
2. Name your project
3. Select target platform(s)
4. Pick a UI framework (XAML or C# Markup)
5. Choose a preset (Blank or Recommended) and Customize
6. Download and open in IDE



What You Get

- Multi-targeted solution structure
- Pre-configured dependency injection
- Navigation setup (Uno.Extensions.Navigation)
- Theming with Material or Fluent styles
- Ready-to-run on all selected platforms

Bridge Design & Code

Seamlessly export Figma designs to XAML or C# Markup

Figma Integration

- **Uno Figma Plugin** exports designs directly to code
- Supports both **XAML** and **C# Markup** output
- Maintains design fidelity – colors, spacing, typography
- Design tokens map to Uno theming resources
- Designers and developers share a single source of truth

Design to Code Workflow

1. Designers create UI in Figma
2. Use the Uno Platform Figma plugin
3. Export selected frames as XAML or C# Markup
4. Import generated code into your project
5. Fine-tune bindings and logic

Visualize Your UI

Real-time visual editing with Hot Design

Hot Design – Uno Platform Studio

- **Live visual editor** for your running app
- Edit UI properties in real time – no recompile
- Drag and drop components
- Inspect the visual tree
- Works alongside Hot Reload for code changes

Hot Design in Action

- Launch your app in debug mode
- Activate Hot Design from the toolbar
- Select elements to edit properties visually
- Changes reflect instantly on all running targets
- Save changes back to your XAML or C# Markup

Go XAML-Free

Build complete UIs using only C# Markup

C# Markup

- Declare UI entirely in C# — no XAML files needed
- Fluent API for building control trees
- Full IntelliSense and compile-time checking
- Strongly-typed bindings
- Same controls and layout system as XAML

C# Markup Example

```
this.DataContext<MainViewModel>((page, vm) => page.Content(
    new StackPanel()
        .Children(
            new TextBlock()
                .Text("Hello, Uno Platform!")
                .FontSize(24)
                .HorizontalAlignment(HorizontalAlignment.Center),
            new Button()
                .Content("Click Me")
                .Command(x => x.Binding(() => vm.OnClick))
        )
    ));
```


XAML vs. C# Markup

Feature	XAML	C# Markup
Syntax	XML-based	Fluent C# API
IntelliSense	Partial	Full
Compile-time safety	Limited	Strong
Hot Reload	Yes	Yes
Figma export	Yes	Yes
Designer support	Yes (Hot Design)	Yes (Hot Design)

Modernize Legacy Apps

Convert WinUI projects to cross-platform apps

WinUI to Uno Migration

- Existing WinUI 3 / Windows App SDK apps can go cross-platform
- **Virtually zero code changes** required
- Uno implements the WinUI API surface on all platforms
- Incremental adoption — migrate one page at a time
- Keep your existing MVVM architecture

Migration Steps

1. Add Uno Platform target frameworks to your project
2. Resolve any Windows-only API usage
3. Use platform-specific code where needed (`#if` directives)
4. Test on each target platform
5. Ship everywhere

Architecture Overview

- **Single multi-targeted project**
 - UI, ViewModels, business logic
- **Uno.WinUI** – cross-platform WinUI API implementation
- **Skia rendering** – pixel-perfect UI via Skia (default on non-Windows)
- **Uno.Extensions** – MVUX, Navigation, DI, HTTP, Auth
- **Uno Toolkit** – pre-built cross-platform UI components
- **MVUX pattern** – Model-View-Update eXtended with immutable records

AI-Assisted Development

Build Uno apps faster with the Uno MCP Servers

Uno MCP Servers

- **Model Context Protocol** servers tailored for Uno Platform development
- Plug into AI assistants like GitHub Copilot, Claude, and Cursor
- Two complementary servers:
 - **Docs MCP** — semantic search over Uno Platform documentation
 - **App MCP** — interact with your running Uno app (inspect visual tree, read state, drive UI)
- Keeps AI grounded in accurate, up-to-date Uno guidance

What You Can Do With It

- Ask "How do I add navigation?" and get answers grounded in current docs
- Generate XAML or C# Markup snippets that actually compile
- Scaffold MVUX models, pages, and bindings
- Let the AI inspect your running app to debug layout or binding issues
- Pair with Hot Design and Hot Reload for a tight feedback loop

Demo Time 🎉

Resources

-  **Uno Platform** — platform.uno
-  **Documentation** — platform.uno/docs
-  **Live Wizard** — platform.uno/create
-  **GitHub** — github.com/unoplatform/uno
-  **Uno Figma Plugin** — Available in the Figma Community
-  **Samples** — github.com/unoplatform/uno.samples

Resources (cont.)

-  **Uno Platform Discord** — Community chat and support
-  **YouTube** — Uno Platform channel for tutorials and walkthroughs
-  **Blog** — platform.uno/blog
-  **Uno Toolkit** — Pre-built UI components
-  **Uno MCP Servers** — AI-assisted development with docs search and app interaction

Thank You!

Questions?

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